

Aitor González-Marfil

Donostia – San Sebastián, Spain (*Open to relocate*) | aitorgacad@gmail.com | linkedin.com/in/aitor-g

Professional Summary

PhD Candidate in CS & AI specializing in **Deep Learning for Biomedical Imaging**. Experienced in developing high-performance Computer Vision (CV) solutions using PyTorch for data-scarce and resource-constrained environments. Bridging academic research and industry application by integrating ML solutions into real-world technical workflows. **Published in Nature** journals with **60+ citations**.

Technical Skills

Programming: Python (expert), Java, C, C++.

Frameworks: PyTorch, TensorFlow, OpenCV, Hugging-Face.

Data & More: NumPy, Pandas, Scikit-learn, Git, Docker, CNNs, Transformers.

Professional Experience

Donostia International Physics Center (DIPC): **ML Engineer** (PhD Research) Mar 2023 - Present

- Developing **PyTorch-based Computer Vision tools** for biomedical imaging, specializing in high-performance solutions for data-scarce and resource-constrained environments.
- Collaborating with biologists to deploy ML models into production-level research workflows.

University of the Basque Country (UPV/EHU): **ML Engineer** (Research) Aug 2021 - Jun 2022

- Leveraged **multitask neural networks** to enhance cross-domain generalization of image models for electron microscopy using TensorFlow.

DIPC & IXA Research group: **Machine Learning Intern** Jun 2020 - Aug 2021

- Biomedical CV:** Implemented Deep Style Transfer for domain adaptation in electron microscopy.
- NLP:** Fine-tuned BERT-based Transformers for multilingual NER and POS tasks using PyTorch.

Education

University of the Basque Country, (UPV/EHU)

PhD in Computer Science	Exp. 2026
M.Sc. Computational Engineering and Intelligent Systems	2022
B.Sc. Computer Science & Artificial Intelligence	2021

Personal Projects

- Light My Cells:** Developed and deployed Dockerized self-supervised Vision Transformer models for fluorescence prediction from label-free microscopy.
- 2048 Bot:** Built a reinforcement learning agent using the NEAT algorithm to play 2048 game.
- Fake News Detector:** Deployed an NLP pipeline using BERT for automated news classification.

Publications

Publications in **Nature Methods** and **Nature Machine Intelligence (60+ citations)**.